

Security Features in Real-World Applications

1. Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:

The security feature/service required, such as confidentiality, integrity, authentication, authorization, availability.

The security mechanism used to provide the identified security feature.

The security protocol or technology used to implement the mechanism.

Briefly explain how the mechanism/protocol provides the identified security feature.

Application	Security Feature	Security Mechanism	Protocol / Technology	How it provides security
WhatsApp	Confidentiality and Authentication	End-to-End Encryption	Signal Protocol (uses Diffie-Hellman key exchange + AES)	It encrypts the messages on the sender's device and only decrypts them on the recipient's device by using unique keys for each session which means that even WhatsApp's servers cannot read the content of the messages.
GitHub	Authentication and Integrity	SSH Key Pairs	SSH (Secure Shell) Protocol	Uses asymmetric key pairs to authenticate a user's machine without transmitting a password and verifies that code pushed to the repository comes from a trusted and unaltered source.

Aadhaar / National ID Verification Systems	Authentication	Biometric Verification	Fingerprint/Iris Matching combined with PKI	This method verifies a person's identity by comparing a physical trait such as a fingerprint or iris pattern against a template stored in the system. Because these traits are unique to each individual and they are far more difficult to steal or replicate than a password.
Microsoft Office 365 (Enterprise Login)	Authentication & Availability	Conditional Access Policies	Azure Active Directory (Azure AD) / OAuth 2.0	The system takes a moment to consider the situation the user is logging in from and where they appear to be located. Whether their device seems trustworthy and how risky the attempt looks overall. If anything about the login seems out of place then access is denied. This way the real users can still get in without trouble while suspicious attempts are stopped before they cause harm.
PayPal	Non-repudiation	Digital Signatures	RSA / PKI	When a payment is made it gets signed using a private key that belongs only to the sender. That signature acts as a kind of proof that the sender genuinely approved the transaction so there is no way for them to later claim they had nothing to do with it.

2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Application	Security Mechanism	Protocol / Technology	Prevention	Detection	Recover
WhatsApp	End-to-end encryption	Signal Protocol (uses Diffie-Hellman key exchange + AES)	The encryption of messages takes place on the sender's device and can only be decrypted at the receiver's end. The traffic cannot be intercepted by WhatsApp's own servers.	WhatsApp warns if user keys change without them changing themselves. It warns if someone is faking their identity.	Keys change constantly so even if one gets stolen the attacker only gets a tiny slice of the chat not the whole history.
GitHub	SSH Key Pairs	SSH Protocol, GPG signing	SSH uses a public/private key pair so no password is ever transmitted and it rules out brute force attacks.	A signed commit lets others check that a code change really came from you and wasn't slipped in by someone else.	GitHub keeps a full history of changes, so a bad or malicious edit can always be traced and rolled back.

Aadhaar / National ID Verification Systems	Biometric Verification	Fingerprint/ Iris Matching combined with PKI	It is harder to fake a fingerprint or iris scan than to guess or steal a password.	Repeated failed match attempts get logged and the system can lock the ID or send it for manual review.	The system keeps the biometric template separate from transaction data so a disputed match can be escalated to human review without ever exposing the raw biometric data.
Microsoft Office 365 (Enterprise Login)	Risk-based Conditional Access evaluated at login	Azure Active Directory (Azure AD) + OAuth 2.0	Suspicious logins, wrong location, unrecognized device gets blocked before they even happen.	The system notices things a real person cannot do like logging in from two countries minutes apart and flags it.	A compromised account can be automatically forced to reset its password or get logged out everywhere.
PayPal	Digital signatures on transactions	RSA / PKI	Every transaction is signed with a private key only the sender has, so nobody else can fake it.	If a transaction gets altered and if the signature no longer matches then the tampering is caught right away.	PayPal keeps signed, timestamped records of every transaction making it easy to investigate disputes and reverse fraud